

Henry Routson

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Extensive knowledge and experience in modern AI systems and Software Engineering. Currently working as an AI Engineer.

Education

The University of Melbourne

Bachelor of Science, Computing and Software Systems
— *Distinction*

The University of Melbourne

Master of Computer Science, AI — *in progress, part time*

- NLP: Fine Tuning, Prompt Tuning, Transformers, Hugging Face

Ballarat Grammar

- Academic scholarship
- VCE: Software Development, Specialist & Methods Mathematics, Chemistry, Physics

Achievements

Antler VC Build Hackathon, 1st Place Jun 2026

antler.co | [Event page](#)

- Won 1st place and \$7,500 prize at Antler's two-day build sprint in Sydney.
- Built [auto-pipeline.dev](#) and [auto-pipeline-app.dev](#) in 14 hours.

Experience

KPMG, AI Engineer Jun 2025 – Present

kpmg.com/au/en.html

- AI Prototype Engineer in Futures, working with LLMs and agent frameworks.
- Developed standardised codebase (34k lines) to automate prompting, parsing, validation, and deep research.
- Automated assessment of 10k pages of legislation equivalent.
- Created pipeline for SQL migration and database merging.
- Deployed Transformer models to CUDA GPUs with Hugging Face and Ollama.
- Built autonomous agent system with RLMs; multi-agent and sub-agent implementation.
- *Skills*: Azure, FastAPI, LangChain, LangExtract, Pandas, Docker.

Bet Right, Data Science Intern Jan 2025 – Apr 2025

[linkedin.com/bet-right](https://linkedin.com/company/bet-right)

- SQL, Snowflake, Databricks, PowerBI, Azure, DevOps.
- Improved abuse detection ML model by 15–40% using stacking.

DSCubed, DS/AI Team Vice Head Nov 2024 – Jul 2025

dscubed.org.au

- RAG, vector databases, LLM frameworks and APIs.
- Contributed to LlamaIndex open-source project.

ARES, University of Melbourne

2023

[linkedin.com/aresunimelb](https://linkedin.com/company/aresunimelb)

- 3D front-end software for rocket trajectory simulation across supersonic and subsonic speeds.

LeetCode: leetcode.com/u/HenryRoutson

Open-Source Contributions

Cached File Explorer (Rust)

[github](#) · [FileExplorer/pull/47](#)

- Ultra-fast file explorer using caching and low-level Rust.
- Fixed a poisoned thread issue from multi-threading

Supabase SQL Auth UI

[github](#) · [flutter-auth-ui/pull/108](#)

- Added options and documentation to the auth UI.

Projects

IT Project (JS/TS, React) dev-aia.vercel.app

- Quicksort visualisation; redesigned stack visualisation and improved code quality.

LLM Experiments

[github/llm_experiments](https://github.com/llm_experiments)

- Transformers, JAX, Keras, PyTorch; combining pre-training and supervised fine-tuning.

Auto Avail (JS/TS)

[github/Auto_Avail](https://github.com/Auto_Avail)

- when2meet-style scheduler using location data to generate continuous availability periods.
- Stack: Supabase, Flutter, React Native, Postgres, pg-TAP, Expo, TypeScript.

CHelp (C)

[github/CHelp](https://github.com/CHelp)

- Valgrind alternative: tracks dynamic allocations and catches memory issues early.

Autoheader (Rust)

[github/autoheader](https://github.com/autoheader)

- Automatically generates C header files.

References

Available upon request.

- **Cindy** — AI Engineer, KPMG. [Linked In](#)
- **Pranav** — L3 Data Scientist, NAB. [Linked In](#)